

System Acceptance Review

BRACU Mongol-Tori

BRAC University

University Rover Challenge – 2019



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1. Introduction

Formed by a group of tech enthusiasts of BRAC University, BRACU Mongol Tori Revolution is the perfect platform for students of different departments to work together and hone their skills. Their collective effort has led to the building of the rover, which is capable of not only assisting astronauts but can also perform science research tasks. Marking the fourth year in the Mongol Tori lineup, the upgraded rover comes with plenty of major changes that makes it more equipped for various complex tasks. The major updates in the current version of the rover include software system, electronics and science module. Moreover, minor changes include such as modifying chassis, improving arm and updating wheel.

The following report address the rover's efficiency as per the requirements of URC 2019 including details on Core Rover Systems, Approach to Competition Tasks, Testing and Operations, Team Budget, Gantt Chart and Science Task. The project was divided into two specific phases- development and testing. The development phase lasted from October'18 till February'19. The testing phase started bit later than development phase from January'19 till February'19.

2. Core Rover Systems: Based on our experience from URC 2018 and after various methods of testing, we have modified our rover to perform at its best. The main focus is to upgrade our rover for maximum performance.

2.1 Mechanical: We have upgraded different aspects of our rover to compete in URC 2019 more profoundly. First of all, we have reduced the height of our rover. This ensures that it is more stable as it has low center of gravity. We have kept our four wheeler design but made an enormous change in our wheel.

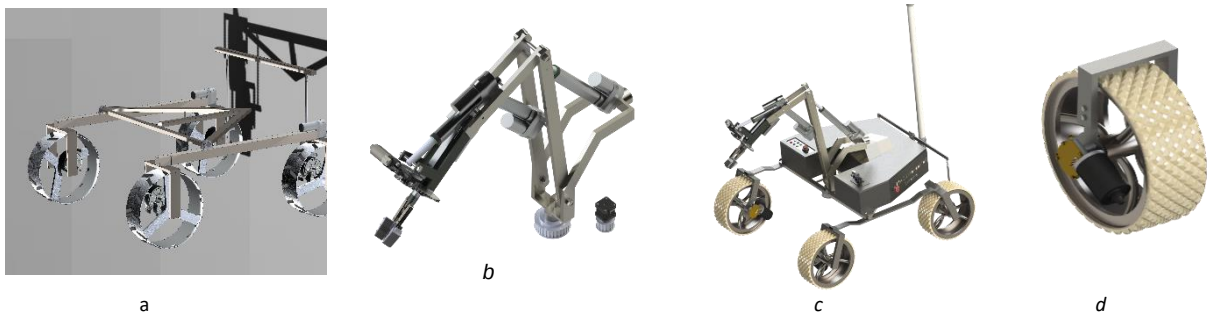


Figure: 1

2.1.1 Chassis and Suspension: Our K shape chassis which is shown in Figure 1(a) is 0.7-meter-long and 0.55-meter-wide and the unique shape has made our rover rigid and firm. Besides, to reduce shape distortion due to unexpected shock from rocky terrain, we have installed triangulation in the chassis. The heavy space frame transmits the flexing load as tension and compression loads along the length of each strut. Moreover, our rover is incorporated with four wheeler modified version of rocker bogie suspension system. In this system, there are two bogies directly connected to the chassis which is supported through a U shaped differential bar at the rear part of the rover. For maximizing the rotational angle of the bogies, we have used universal joints in the differential bar. The unique feature of this system is the ability of expansion of the bogies. The length of the bogies turns to 1.2 m from 0.76 m using two heavy duty linear actuators. The expanded suspension provides more stability during the vertical drops and slopes. Figure 1(c) shows our full working rover's design.

2.1.2 Wheel: After our experience in URC 2018, improving the wheel's design was one of the major priorities. We have improved many aspects of the design of our wheel to ensure smooth movement of our rover. This year our wheel has 0.3-meter diameter and 0.10-meter width. As shown in Figure 1(d) the wheel is made by stainless steel so that the wheels are lightweight and difficult to break. We have modified our wheel using U shaped rim joined with stainless steel ring. This design has increased shock

absorbing capability of the rover. Moreover, we drilled small holes in our wheel to make it lighter. For rectifying 360-degree rotation, climbing rock and vertical drop, rubber pad is used on the wheel so that the wheel can have better grip.

2.1.3 Arm: We have configured our rover with a strong robotic arm having six-degree of freedom (base, shoulder, elbow, wrist, two degree of freedom in the end effector). The arm shown in Figure 1(b) consists of four linear actuators and two dc motors. Gear mechanism has been assimilated at the tip of the arm, which gives the end effector more degrees of rotational freedom without wire twisting. A linear actuator system in the claw gives superior gripping flexibility. We are using three different arms which are light weight, strong and stable. They are suited for different tasks. Our main focus is strength and stability for precise movements of the arm. We have used motor and pulley gear in the base for rotation and stability. Moreover, 3 feedback end effectors have been incorporated with the arm to increase our precision while performing different tasks. Along with that, 3 different claws are configured with the robotic arm for different tasks. We are using a 2 finger claw for grabbing small parts and equipment servicing. A heavy claw is configured which is capable of lifting more than 5 kg for performing heavy tasks in equipment servicing and extreme retrieval. There is also a soil sample collector claw.

2.2 Communication system:

We have taken a number of measures in order to create a seamless flow of data to and from the rover and base station. We have used two routers- Rocket M2 in the base station and Bullet M2 on the rover- for mainly transferring video and control signals. A NUC mini PC is installed on the rover to communicate with the base station via the routers. Moreover, an omnidirectional antenna (TL-ANT2412D) has been fitted on the rover, paired with the Bullet M2 router to attain a better coverage due to its 360-degree horizontal beam pattern. On the other hand, a high gain directional antenna (ANT232D15T-120DP) has been mounted on the base station which mainly ensures communication with the rover while also tracking it using a rotator (G-5500). The rotator itself is controlled using an elevation azimuth controller based on the rover's movement and corresponding level of signal strength. In order to maintain a clear view of the rover's surroundings, three IP cameras along with six FPV cameras have been employed. We also have three monitors (3 monitors and 1 VR set) using which we can switch between the streams from the six FPV cameras, from the onboard control software. We use UDP protocol as it allows fast capture and transmission of video data, as well as TCP protocol for control signal transmission as it allows better reliability. In case of video streams from FPV camera, 5.8 GHz frequency band is used, while for control signals and IP camera output, a 2.4 GHz band is used.

2.3 Control system:

We have built a portable base station which will be operated with 1 laptop and 3 monitors for camera view as well as 3 joysticks for smooth controlling of the rover.

2.3.1 Electronics: This year, we have upgraded our electronics sector using various new components. The components were selected based on previous experience and different test run. The total configuration is shown in Figure 2(a). After analyzing performance, we have used high torque DC Geared motor. We are using 10,000mah Li-Po battery which will supply power for approximately 42 minutes. Our motor combination provides average rated current of 10 amperes, average current of 14 ampere and stall current of 20 amperes. In addition, we are using four 30 ampere Cytron motor driver (MD30C) for wheels and six 10 ampere Cytron motor driver (MD10C) for arm. We are using ATmega 2560 for controlling wheels and arm. To supply power to the wheel we are using 2 Li-Po batteries. Proteus and Eagle have been used for PCB design. For science task, we are using two (1p2s) li-ion and one (2p3s) li-ion battery pack. Besides, we are using a 24V Li-Po battery for distributing power to 24V router, 12V NUC and 5V

switch for communication purpose. Finally, we are using a high ampere kill switch for stopping our rover in emergency situation.

2.3.2 Software: We have built up a few graphical UIs for control, science and mapping with an advantageous programming framework. The mapping GUI tracks the rover continuously on a pre-loaded offline map using the latitude and longitude received from the GPS sensor. We have built up the control GUI shown in Figure 2(b) using JAVA programming language. This framework follows the Transmission Control Protocol (TCP) and the base station acts the client while the rover acts as the server. Furthermore, we have built up a feedback system which is one of the vital upgrades in our software system. We have implemented feedback receiving system in our control GUI so that we can use feedback end effectors along with feedback mechanism. In addition, we have utilized C# to make the science GUI which deciphers the sensor readings into custom measures and offer a graphical recognition that makes it simple to recognize distinctive conditions of observations. We have made a control programming shown in Figure 2(c) for on board science test in our science GUI. The autonomous segment of the software is built by using C++ programming language and computer vision algorithms. The software combines both GPS and magnetometer values in coordination with the visual data from the camera to achieve full autonomy.

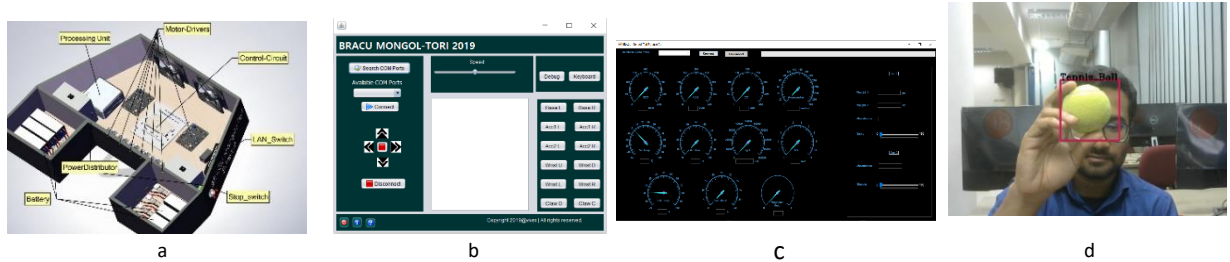


Figure: 2

3. Approach to Competition Missions: We have arranged a mock competition setup similar to URC to practice and complete all the task in time. The experience of URC 2018 helped us to select locations for practice, where all the tasks are strictly scheduled to be performed in a time limit.

3.1 Extreme Retrieval and Delivery Mission: Our rover has been designed in a way that have the ability to travel up to 1km. The modified rocker bogie suspension and upgraded chassis is suitable for performing Extreme Retrieval and Delivery Mission. Our suspension system is capable of absorbing sudden shock while moving through rough terrain. During the vertical drop, our expanded rover body allows the rover to land firmly. Our heavy arm is capable of lifting heavy weight materials such as tool box or water bottles.

3.2 Equipment Servicing Mission: Equipment servicing requires precise and strong movement. To ensure precise movement to operate switch and keyboard, we are using 3 finger claw. Moreover, to ensure precise movement, we are using feedback end effector for arm's precision and angle calculation. The

formula is: $\theta_x = \text{ArCoS} \frac{x^2 + y^2 - \alpha_1^2}{2xy}$.

Previously, a camera was used to decide arm movement but this year, we have shifted to angle calculation. This not only determines the movement that occurred and also the movement that is necessary.

3.3 Autonomous Traversal Mission: In order to achieve a seamless autonomous traversal, a combination of sensors, cameras and computer vision techniques have been employed- a Sparkfun's Venus GPS board based on Venus638FLPx GPS module and a HMC6343 three-axis magneto-resistive module have been used. Using the GPS module, we calculate the latitude and longitude (with a 5 meters' error margin) from an open environment with a preloaded map. The autonomous system works based on computer vision processor receives information from GPS, compass and the cameras. For locating our rover in real time, we have created a custom made offline map using python and geo referencing map from open street map. Once the rover is within 3.5 meters' radius of the destination (as detected by the GPS module) the image processing sub-routine takes over. The routine has been constructed using dark flow, TensorFlow, OpenCV and yolo version 2 with an average accuracy of 81% based on our custom model and dataset to

detect the tennis ball. After receiving the coordinates from the GPS sensor, the sub-routine calculates the distance to the destination and the rover autonomously traverses the path accordingly. In addition, a magnetometer is employed whose data is used by the software to keep the rover aligned in the direction of the determined travel path.

4. Safety measures: While developing the rover we have strictly followed the safety purpose. Our rover has a “kill switch” which will directly cut off all the power from the mechanism while needed. For ensuring more safety we have installed a 30amp fuse box for the main circuit board.

5. Testing and Operations: Several experiments were conducted prior to the competition to ensure proper functionality of the rover. The experiments can be split into three steps:

a. Sub-System Testing:

The subsystems include wheels, suspension & chassis, end effector, sample collector, on-board science test and software. Before implementing the design, we have tested all the mechanism in solid works and modified our designs from 17/10/2018 to 01/01/2019. On 12/01/2019 we made a custom ramp for testing different types of wheels. After selecting the wheel, we had our indoor test on our rover chassis and suspension test on 14/01/2019. Considering different types of gear mechanisms, we have selected worm gear assembly for our end effector and tested the module on 15/01/2018 by grabbing small objects such as screwdriver, hammer etc. in our lab. Since we got a good result in URC-17 & 18 by using the drill mechanism for sample collection, this year we are considering a new excavator claw mechanism alongside a modified version of the drill mechanism. To bring some improvement we will construct the drill mechanism after SAR and by this time we have conducted a test of the excavator claw on 20/01/2019 and tested outside on 27/01/2019. Moreover, we have developed a platform independent control GUI which is compatible with any system and rover. By getting this advantage we have tested our control GUI on a prototype rover on 02/02/2019.

b. Complete System Testing:

The communication and electronics system used in URC-2018 performed very well so we have decided to build upon the same systems. For getting a positive confirmation with the new updated communication and electronics system the full setup was tested on the rover. It was done at the very beginning of our work for URC-2019 as well as in the last of January 2019. After preparing the total robotic arm system and different types of end effectors we have tested the full arm system on 18/02/2019 by picking up different objects and collecting soils.

c. Operational Testing

For operational testing purpose of our complete rover system we focused on the main four tasks of URC-2019. We have chosen a remote place where the surface was quite similar to the main surface of competition ground. In this place we had a point for vertical drop, steep slope and different types of rough terrain. We went to this test site for extreme retrieval task with our rover and started the testing from January 2019. For testing the equipment servicing task, we have made a dummy setup and tested our rover inside our laboratory on 02/02/2019. In this test we have practiced switch on-off, turning knob, collecting cache box and undoing a latch etc. For the autonomous test, we chose another location where the GPS data were available. For a complete test of autonomous task on 11/02/2019 we have selected some points on the basis of GPS values and successfully completed the task. While we were doing our extreme retrieval and delivery task, we collected some soil with our rovers' sample collector to perform the science task and on-board tests in our rover's science module part. Regarding the science module, on 18/02/2019 we have tested the soil for traces of life and we have successfully done microbial biomass test, water flow capillary test and amino acid test.

Science Task:

In Science Task to convert our rover into automatic planet analyzer, we have divided our plan into three segments which are 1. Digging module and sample collector, 2. Environmental data collecting sensor box, 3. Automatic on board laboratory. We have designed an arm with six degrees of freedom with an excavator claw which can dig through hard soil surface & can go up to 20 cm or more. It can collect the soil & put it in sample collecting box. At once it can supply up to 80 gm of soil. The on-board sensor box is equipped with Air Temperature Sensor (DHT22), Air Humidity Sensor (DHT22), Soil Temperature Sensor (DS18B20), Soil Moisture Sensor (DS18B20), CO Sensor (MQ7), CO2 Sensor (MQ135), H₂ Sensor (MQ8), O₂ Sensor (GROVE gas sensor), Light Intensity Measurement Sensor (LDR), UV Light & radiation Measurement Sensor (ML8511), Hall Effect Sensor for magnetic field detection, Compass Sensor for Navigation, Barometric pressure sensor, Anemometer for determine the speed wind blow of an area, 360 degree Panorama View of Arena, pH Soil Probe Sensor, and a endoscopic camera which can take clear picture of the trench. From those result we can get a decent idea of planetary environmental data, climate, soil composition and formation of landscape. Among the sensors, soil temperature, soil moisture, pH probe are connected at the top of the arm shown in Figure 4(a). In order to implement automatic on board laboratory test, we ensured analyzing of soil sample & finding bio signatures or any possibilities of forming life in future and water history. To do so, the on board laboratory is divided into 2 boxes which can perform 5 tests precisely. In box 1 we have performed Biomass test, Amino Acid test and Water Flow Capillary test. In Biomass test we have used load cell to detect weight variation and a nichrome wired heater for heat generation. The formula is: **Biomass = (difference of weight/previous weight) * 100**. In Amino Acid test, along with a nichrome wired heat generator and test tube, we have used ninhydrin reagent to detect the present of amino acid in the sample. Moreover, a centrifugal tube is used in Water Flow Capillary test to hold soil & a servo motor pushes a syringe which can supply water in the tube. It provides the water absorption rate of soil at that area. This test indicates whether the soil surface is able to hold the water on surface as surface water or absorb and store it as ground water. A sample collecting chamber is incorporated to store pure form of sample for future experiment. Box 2 performs Spectroscopy test which focuses on performing quantitative analysis of sample by using spectroscopic method. There is a light source and a sensor which can detect the light absorbance. By this we can measure the amount of Nitrogen, potassium & phosphorus which are very necessary elements for both soil properties & biological activities. Last test is Microscopy of soil composition test in which a digital microscope is mounted in this box with the zooming capability up to 800 times and provide information of soil composition that indicates soil type.

There is one central soil distribution system for each boxes. It is custom made and fully 3D printed which distributes sample to the located site of that box according to their procedure.

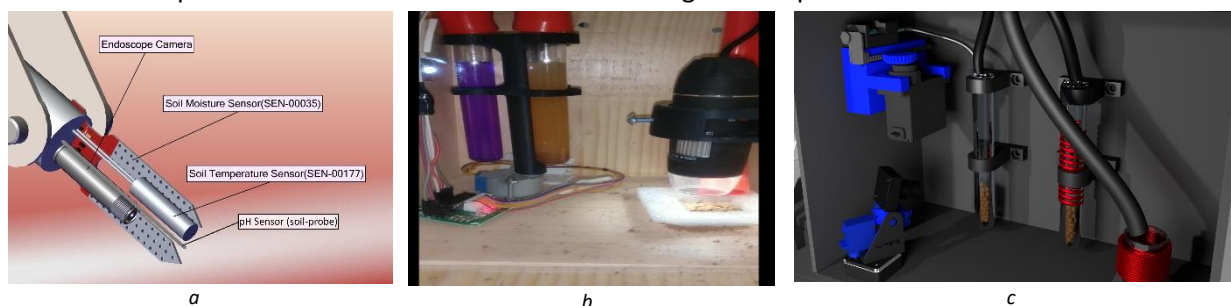


Figure: 4

Reference: NASA's Mars Exploration Program's Science Theme. (n.d.). <https://marsmobile.jpl.nasa.gov/programmissions/science/>
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